

Implementing a Master-Slave J-K Flip Flop

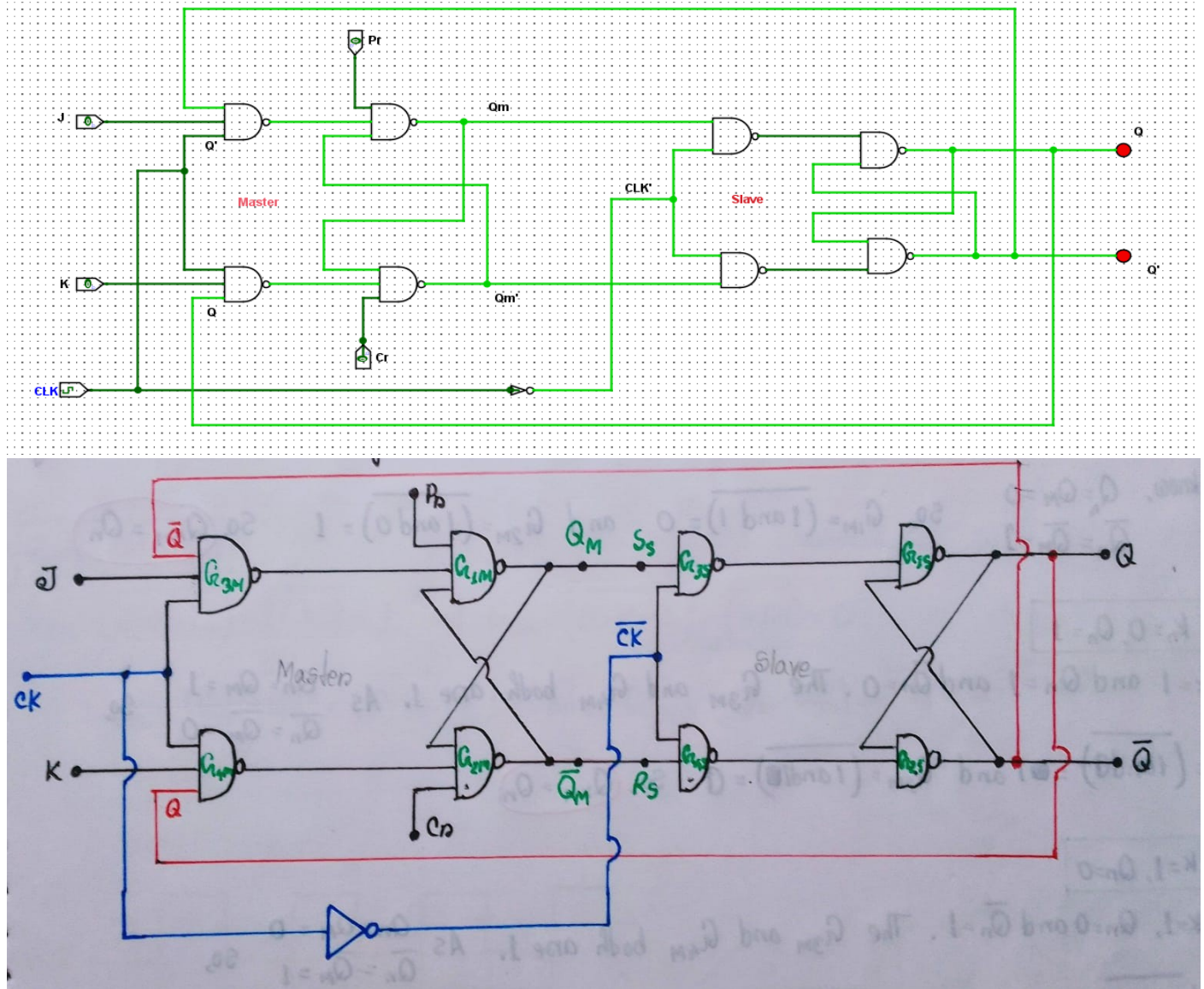


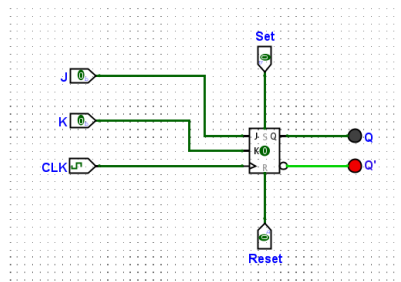
Figure 1: NAND Gate Implementation of a Master-Slave J-K Flip Flop

```

1 timescale 1ns/1ps
2
3 // =====
4 // JK Master-Slave Flip-Flop - Gate-Level Structural Model
5 // Equivalent to 7476 IC behavior
6 // Trigger: Falling edge of CLK
7 // PRE_bar / CLR_bar: Active LOW asynchronous override
8 // =====
9
10 module jk_ms_ff (
11     input j,
12     input k,
13     input clk,
14     input pre_bar, // Active LOW async preset - forces Q=1
15     input clr_bar, // Active LOW async clear - forces Q=0
16     output q,
17     output q_bar
18 );
19
20 wire clk_bar;
21 wire m1, m2;
22 wire s1, s2;
23 wire s1_bar, s2_bar;
24
25 // Clock inversion - slave is active when clk_bar=1
26 not #1 clk_bar (clk_bar, clk);
27
28 // Master SR latch - pre/clr included to break x on startup
29 nand #1 g3m (m1, j, q_bar, clk);
30 nand #1 g4m (m2, k, q, clk);
31
32 // Master SR latch - pre/clr included to break x on startup
33 nand #1 g1m (s1, s1_bar, pre_bar);
34 nand #1 g2m (s2, s2_bar, clr_bar);
35
36 // Slave steering gates - active only when clk_bar=1 (CLK=0)
37 nand #1 g3s (s1, s1_bar, clk_bar);
38 nand #1 g4s (s2, s2_bar, clk_bar);
39
40 // Slave SR latch - true async pre/clr override at output
41 nand #1 g1s (q, s1, q_bar, pre_bar);
42 nand #1 g2s (q_bar, s2, q, clr_bar);
43
44 endmodule
45
46 module tb_jk_ms_ff;
47     reg j, k, clk;
48     reg pre_bar, clr_bar;
49     wire q, q_bar;
50
51     jk_ms_ff uut (
52         j, k, clk,
53         pre_bar, clr_bar,
54         q, q_bar
55     );
56
57     // Clock: 20ns period
58     initial clk = 0;
59     always #10 clk = ~clk;
60
61     // VCD dump for GTKWave
62     initial begin
63         $dumpfile("jk_ms_ff.vcd");
64         $dumpvars(0, tb_jk_ms_ff);
65     end
66
67 // Truth Table
68 // =====
69 initial begin
70     pre_bar = 1; clr_bar = 1; j = 0; k = 0;
71
72     // Hold clr_bar low for full clock cycle to flush x from both latches
73     clr_bar = 0; #30; clr_bar = 1;
74     #10;
75     @(negedge clk);
76
77     $display("");
78     $display(" J K | Q Q_bar | Operation");
79     $display("-----");
80
81     // HOLD (Q was 0)
82     j=0; k=0; @(negedge clk); #2;
83     $display(" %b %b | %b %b | HOLD", j, k, q, q_bar);
84
85     // SET
86     j=1; k=0; @(negedge clk); #2;
87     $display(" %b %b | %b %b | SET", j, k, q, q_bar);
88
89     // HOLD (Q was 1)
90     j=0; k=0; @(negedge clk); #2;
91     $display(" %b %b | %b %b | HOLD", j, k, q, q_bar);
92
93     // RESET
94     j=0; k=1; @(negedge clk); #2;
95     $display(" %b %b | %b %b | RESET", j, k, q, q_bar);
96
97     // TOGGLE (0 -> 1)
98     j=1; k=1; @(negedge clk); #2;
99     $display(" %b %b | %b %b | TOGGLE", j, k, q, q_bar);
100
101     // TOGGLE (1 -> 0)
102     j=1; k=1; @(negedge clk); #2;
103     $display(" %b %b | %b %b | TOGGLE", j, k, q, q_bar);
104
105     $display("-----");
106     $display("");
107
108     // Async PRE test
109     pre_bar = 0; #15;
110     $display(" PRE_bar=0 -> Q=%b Q_bar=%b (expect Q=1)", q, q_bar);
111     pre_bar = 1; #20;
112
113     // Async CLR test
114     clr_bar = 0; #15;
115     $display(" CLR_bar=0 -> Q=%b Q_bar=%b (expect Q=0)", q, q_bar);
116     clr_bar = 1; #20;
117     $display("");
118 end
119
120 // Random Sequence for GTKWave
121 // =====
122 integer seed = 42;
123 integer i;
124
125 initial begin
126     $display(" J K | Q Q_bar | (random sequence)");
127     $display("-----");
128
129     for (i = 0; i < 20; i = i + 1) begin
130         @(negedge clk);
131         j = $random(seed) % 2;
132         k = $random(seed) % 2;
133         @(negedge clk); #2;
134         $display(" %b %b | %b %b | ", j, k, q, q_bar);
135     end
136
137     $display("-----");
138     $display("Done. Open jk_ms_ff.vcd in GTKWave.");
139     $finish;
140 end
141 endmodule

```

Figure 2: Verilog script for the Master slave JK Flip Flop



J	K	Q	Q_bar	Operation
0	0	0	1	HOLD
1	0	0	1	SET
0	0	1	0	HOLD
0	1	1	0	RESET
1	1	0	1	TOGGLE
1	1	1	0	TOGGLE

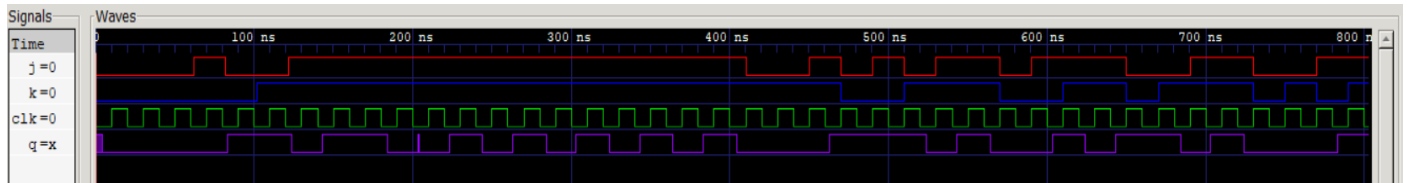


Figure 3: Block Diagram, Truth Table and Signal Response for the Master Slave J-K Flip Flop